Bridging the Gap

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A New Approach to Infrastructure Modeling and Visualization

Introduction & Our Motivation

There are over 617,000 bridges in the United States. Each is paramount to ensuring the reliable and efficient transit of goods, services, and individuals. Despite each bridge's critical function, a large portion are in a state of disrepair; this issue is not well-recognized in part due to the lack of non-proprietary solutions for transparent exploration of the data. Unfortunately, the most recognized tool for bridge data exploration, "InfoBridge," lacks the clarity and flexibility expected of modern apps. This gap in technology motivated us to build a user-friendly application to help others better understand existing bridge data and modern deterioration modeling approaches. As such, we combined weather, traffic, and bridge data to build a modeling and visualization framework that improves upon current solutions. This new tool takes the form of an interactive, Flask-based web application that offers simple access to bridge infrastructure data, visualizations, and predictive models.

Data Approach & Unification

Our application is supported by three primary data sets: bridge information from the Federal Highway Administration spanning back to 1993, monthly historical weather data from 1990 to Present from the National Oceanic and Atmospheric Administration, and monthly traffic volume data from 2002 to 2020 from the Federal Highway Administration. After being downloaded manually, these datasets were cleaned and loaded into our AWS MySQL database via Python scripts. Using cross-reference tables, we joined these datasets together based on state name, nearest weather station, or other fields. Bridge condition predictions were loaded into MySQL via Python and joined via the unique structure number of each bridge. In total, this data amounts to 6,509,127 rows or roughly 1.9 GiB of disk space.

Modeling Experiments & Evaluation

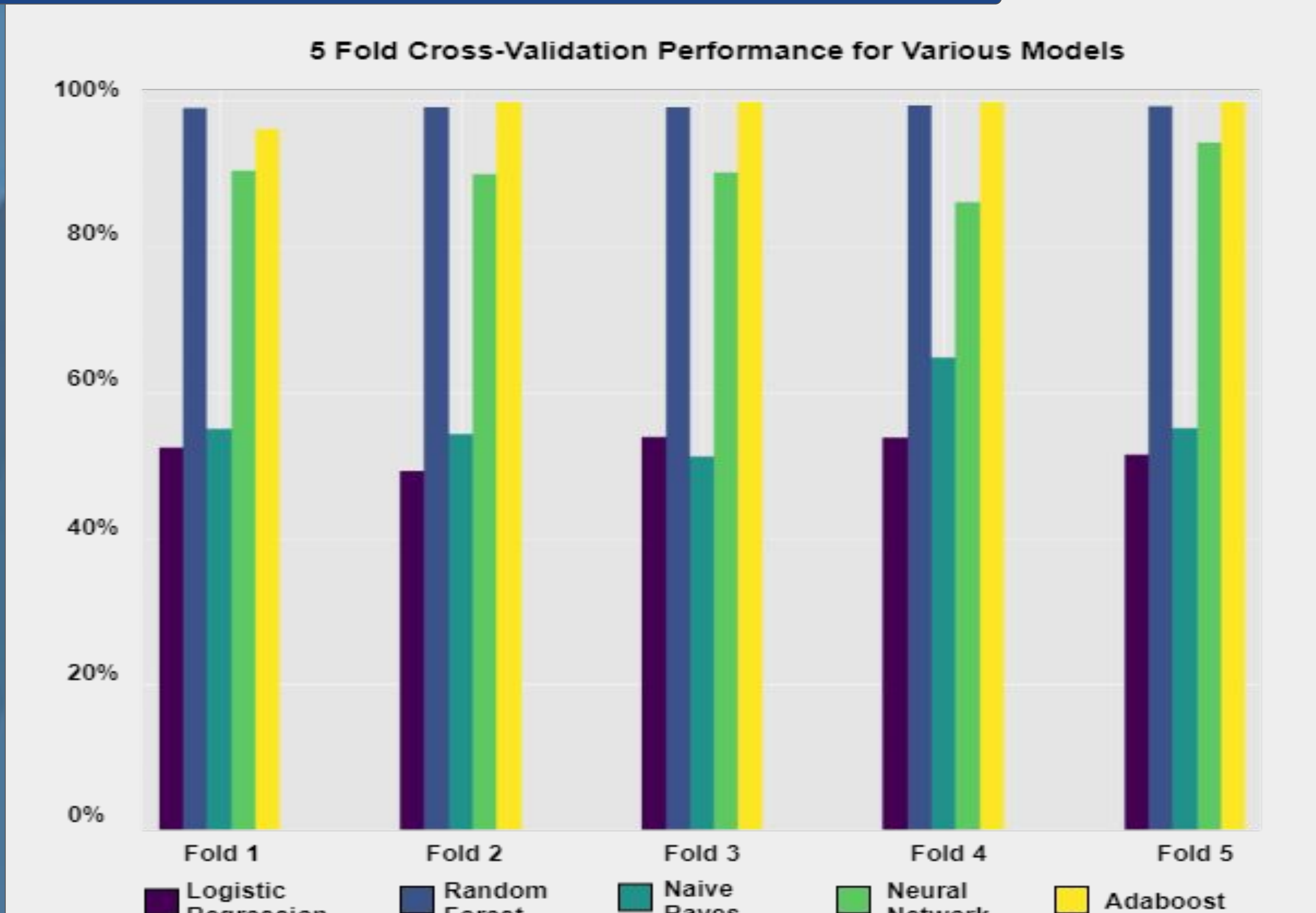

Figure 1: Cross-Validation Accuracy of Each of the 5 Model Classes Tested on the Southeast U.S.

Through 5-fold cross-validation and subsequent hyper-parameter optimization of five leading modeling approaches for the categorical prediction task of future bridge condition (Poor, Fair, or Good), we were able to achieve results that significantly improve upon the present state-of-the-art models in three key aspects. Firstly, unlike the existing body of academic research on the task, our models leverage additional and novel data sources such as traffic volumes and weather. Secondly, our models generalize beyond a single state's bridge inventory which most current research is limited to and instead tackles the problem at a regional level. Finally, our production random forest and AdaBoost models significantly improved the existing benchmark categorical accuracy metric achieved by Heng Liu and Yunfeng Zhang [1] using convolutional neural networks from 85% to 99.24% and 99.16% respectively. Additionally, neural network models improved upon this existing accuracy benchmark to 94.32%. As shown in the cross-validation performance plot (Figure 1), the modeling performance of more complex model architectures such as random forest, adaboost, and neural networks proved far superior to more interpretable, but less complex approaches such as naive Bayes and logistic regression.

Visualization Approach & Evaluation

We used Tableau to display the data, utilizing Tableau Public's self-hosting option to publish the dashboards and embed them within our Flask application. The first of two dashboards is a detailed choropleth map plot of individual bridges; this plot allows the user to view each bridge's location, average traffic in the state, average precipitation from the nearest weather station, bridge building material, bridge condition model predictions of each bridge for the previous year, current year, and next year, as well as other important information. The second dashboard is a higher-level view that allows the user to interact with the model results; users can select a prediction year and a model type, allowing them to view how the predictions change for dimensions such as bridge class and structure type. Displaying the data through these visualizations allows users to easily navigate a complex data set consisting of over 600,000 bridges and intuitively view key features about those bridges. Our approach improves upon the existing solutions by providing an easy-to-navigate interface and incorporating traffic and weather data to provide a more comprehensive view of U.S. bridges.

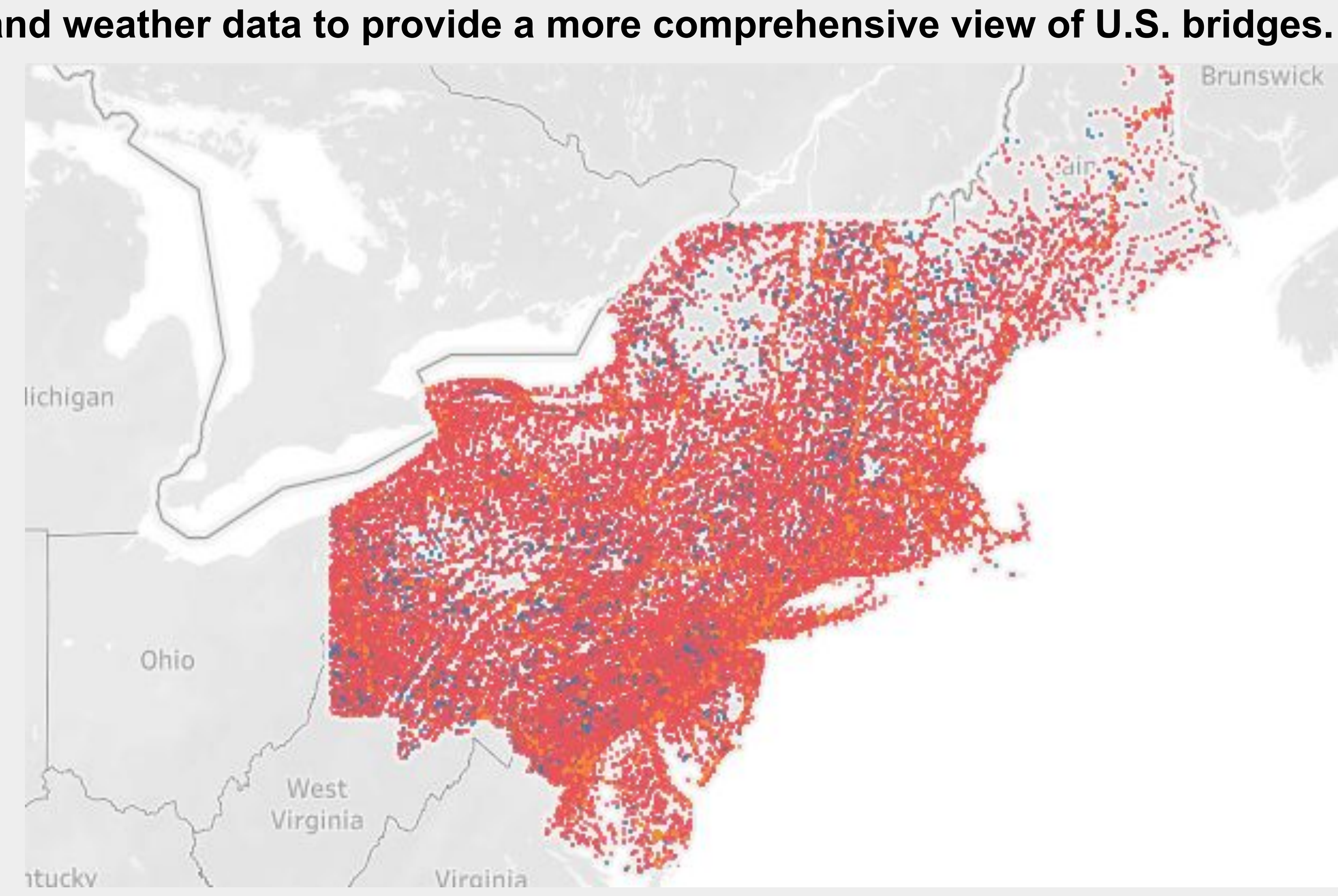

Figure 2: Example of Northeastern U.S. Bridge Data from our Dashboard

To evaluate our new visualization approach, we had 10 users compare the existing InfoBridge portal and our Flask app. These users were asked to rate each app across qualities including clarity, ease-of-use, and functionality. Users rated the Flask app an average of 4.5 out of 5, with the highest marks being in functionality and ease-of-use. The InfoBridge portal was rated an average of 2.5 out of 5, primarily due to low marks in functionality.

Infrastructure Approach

The figure below shows the application's infrastructure. Using an HTTP request in a web browser, users are immediately served our Flask application hosted in AWS. This application serves embedded Tableau visualizations connected to data in our MySQL database and static information connected via an AWS S3 bucket. By using modern and primarily open-source tools, our application can focus on responsiveness, ease-of-use, and interactivity. As was detailed in our user study (mentioned above), our approach creates a better overall experience compared to existing tools.

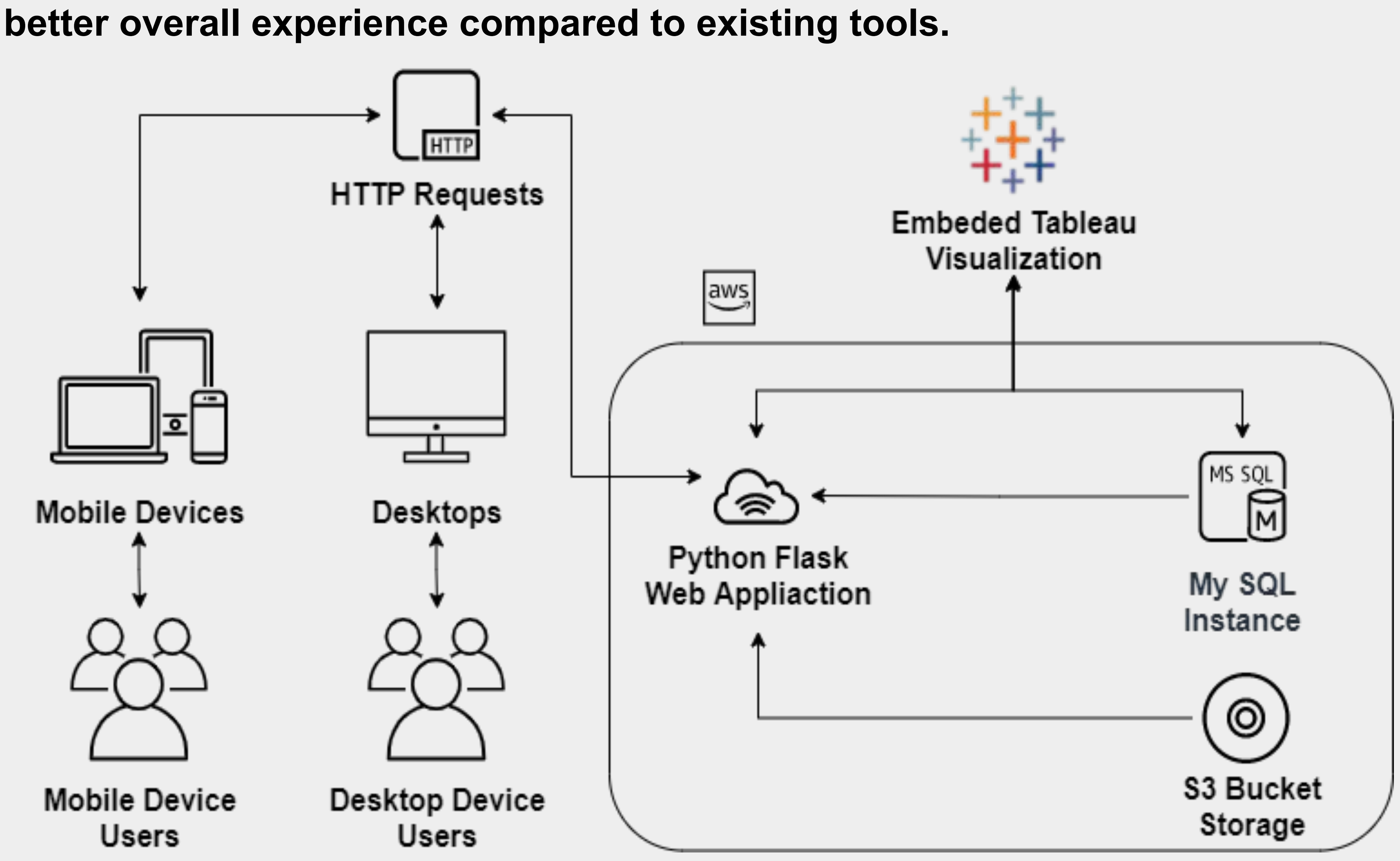

Figure 3: Diagram of the Application's Infrastructure

[1] H. Liu and Y. Zhang, "Bridge condition rating data modeling using deep learning algorithm," Structure and Infrastructure Engineering, vol. 16, no. 10, pp. 1447–1460, 2020. [Online].Available: https://doi.org/10.1080/15732479.2020.1712610